CHEMICAL THERMODYNAMICS · CLASS NOTES

Heat Capacity, P–V Work & Enthalpy

Nine Education IIT Academy · Class 11 Chemistry

1 Heat Capacity

Everyday observation: supply more heat to a body and its temperature climbs further; remove more heat and it falls further. The heat exchanged is therefore proportional to the temperature change:

$$q \propto \Delta T \quad \Rightarrow \quad q = (\text{constant}) \times \Delta T \quad \Rightarrow \quad \Delta T = \frac{q}{\text{constant}}$$

The constant depends on the body. Put the same amount of heat q into a pan of oil and a pan of water and the two temperatures rise by different amounts, because the two liquids have different constants. This constant is called the **heat capacity**, C.

◆ **DEFINITION**

Heat capacity is the quantity of heat needed to change the temperature of a given body by 1 K (equivalently 1 °C).

$$q = C\,\Delta T \qquad C = \frac{q}{\Delta T} \qquad \text{unit: } \mathrm{J\,K^{-1}}$$

Three ways of expressing it

Type	Heat needed to raise…	Defining equation	Unit
Total heat capacity (C_T)	the whole given sample by 1 K	$C_\mathrm{T} = \frac{q}{\Delta T}$	$\mathrm{J\,K^{-1}}$
Molar heat capacity (C_m)	1 mole of substance by 1 K	$C_\mathrm{m} = \frac{q}{n\,\Delta T}$, so $q = nC_\mathrm{m}\,\Delta T$	$\mathrm{J\,mol^{-1}\,K^{-1}}$
Specific heat capacity (c)	1 g of substance by 1 K	$c = \frac{q}{m\,\Delta T}$, so $q = mc\,\Delta T$	$\mathrm{J\,g^{-1}\,K^{-1}}$

Link between them: $C_\mathrm{T} = n\,C_\mathrm{m} = m\,c$

2 Heat Capacity in Different Processes

Heat capacity is **not a fixed number** for a system; it depends on the **path** followed while heat is supplied. Using $C = q/\Delta T$:

Process	What happens	Heat capacity
Isothermal	$\Delta T = 0$ while $q \neq 0$, so $q/\Delta T$ blows up	$C = \infty$
Adiabatic	$q = 0$ while $\Delta T \neq 0$	$C = 0$
Isobaric (constant P)	heat added at fixed pressure	$C = C_p$
Isochoric (constant V)	heat added at fixed volume	$C = C_v$

3 Worked Example: Heat Capacity & the First Law

EXAMPLE 1 **Finding ΔU from a path-dependent heat capacity**

Problem. An ideal gas expands and does 100 J of work on its surroundings. For this process the heat capacity is $+2\ \text{J}/^\circ\text{C}$. If the final temperature is 25 °C above the initial temperature, find the change in internal energy of the gas.

STEP 1 Data (chemistry sign convention: work done *by* the gas is negative).

$$w = -100\ \text{J}, \qquad C = 2\ \text{J}/^\circ\text{C}, \qquad \Delta T = +25\ ^\circ\text{C}$$

STEP 2 Heat absorbed along this path.

$$q = C\,\Delta T = 2 \times 25 = +50\ \text{J}$$

STEP 3 First law.

$$\Delta U = q + w = 50 + (-100) = \boxed{-50\ \text{J}}$$

Interpretation: the gas spent more energy as work (100 J) than it received as heat (50 J), so its internal energy drops by 50 J.

4 Work

Work is energy passed from one system to another by a means in which a temperature difference is not directly involved. Work is a **path function**: its value depends on how the change is carried out, not only on the initial and final states.

Units

- Heat and work are both forms of energy, so both are measured in **joules (J)** in SI.
- A lot of older data is quoted in **calories (cal)**.
- The product $P \times V$ (litre × atmosphere) also has dimensions of energy, so it is handy to memorise the conversion:

MEMORISE

$$1\ \text{L atm} = 101.3\ \text{J}$$

5 P–V Work

Take a gas in a cylinder closed by a frictionless piston of area A. An external pressure P_{ext} pushes the piston through a small distance dx. Work = force × displacement:

$$dw = F\,dx = (P_{\text{ext}}\,A)\,dx = P_{\text{ext}}\,dV \quad \text{(magnitude)}$$

Attaching the sign so that work done **on** the system is positive:

KEY RESULT

$$w = -P_{\text{ext}}\,\Delta V = -P_{\text{ext}}\,(V_2 - V_1)$$

Change	ΔV	Sign of *w*	Meaning
Expansion	positive	negative	system **loses** energy as work
Compression	negative	positive	system **gains** energy as work

6 Work in Isothermal Processes

6.1 Reversible isothermal expansion

Here T is constant and the process is carried out so slowly that the gas is always in equilibrium with its surroundings, so $P_{\text{ext}} \approx P_{\text{gas}} = nRT/V$ at every instant.

$$dw = -P_{\text{ext}}\,dV = -\frac{nRT}{V}\,dV$$

Integrating from V_1 to V_2 (T constant):

KEY RESULT

$$w_{\text{rev}} = -nRT\ln\frac{V_2}{V_1} = -2.303\,nRT\log_{10}\frac{V_2}{V_1}$$

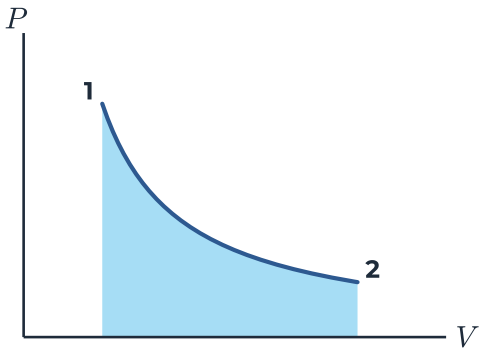


Fig. 1 – *Reversible isothermal expansion (1 → 2); shaded area = $|w_{\text{rev}}|$*

Graphically, the magnitude of the work equals the **area under the P–V curve** between V_1 and V_2.

6.2 Irreversible isothermal expansion

The gas expands against a **single constant external pressure** equal to the final gas pressure P_2. The area under the path is now a simple rectangle:

$$w_{\text{irr}} = -P_{\text{ext}}(V_2 - V_1), \qquad P_{\text{ext}} = P_2$$

Because this rectangle lies *inside* the area under the reversible curve, for expansion:

REMEMBER

$$|w_{\text{rev}}| > |w_{\text{irr}}|$$

The gas does the **maximum work** when expanding reversibly.

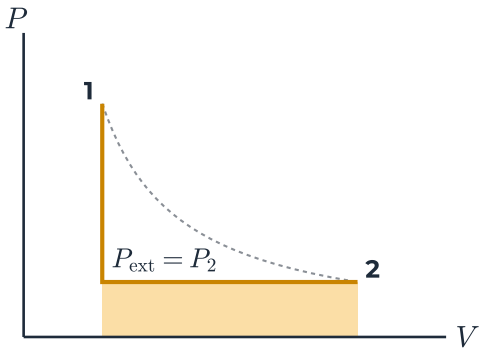


Fig. 2 – *Irreversible expansion against constant* $P_{\text{ext}} = P_2$

6.3 Reversible isothermal compression

Run the same curve backwards: the gas now goes from state 1 (low pressure, large volume) to state 2 (high pressure, small volume). The work is again the area under the curve, now **positive** since work is done **on** the gas.

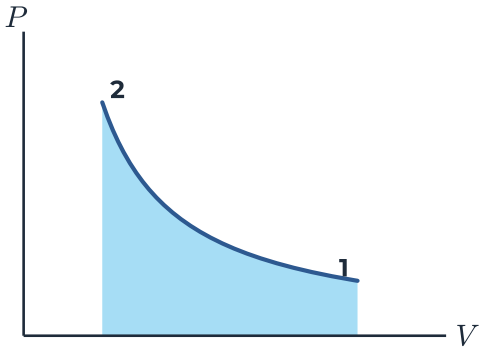


Fig. 3 – *Reversible isothermal compression (1 → 2); shaded area =* w_{rev}

6.4 Irreversible isothermal compression

The external pressure is raised **suddenly** to the final pressure and held constant while the gas is squeezed:

$$w = -P_{\text{ext}}(V_2 - V_1)$$

positive, since $V_2 < V_1$

The rectangle now *contains* the area under the reversible curve, so for compression:

REMEMBER

$$w_{\text{irr}} > w_{\text{rev}}$$

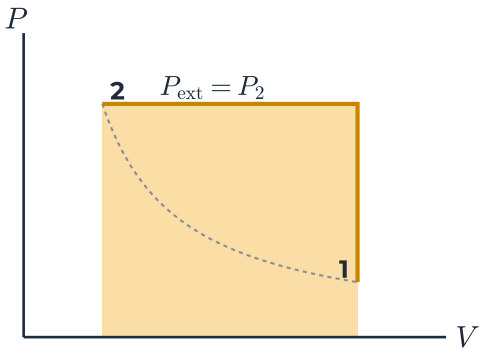


Fig. 4 – *Irreversible compression at constant* $P_{\text{ext}} = P_2$

◆ **CONCLUSIONS TO REMEMBER**

- For a given change of state, the work **done on the gas** is **smallest** when the compression is **reversible**.
- For a given change of state, the work **done by the gas** is **largest** when the expansion is **reversible**.
- Expansion or compression against a **constant external pressure** is **irreversible**.
- Any **sudden** change in conditions makes the process **irreversible**.

EXAMPLE 2 **One gas, two paths**

Problem. One mole of an ideal gas at 300 K expands isothermally from 10 L to 20 L. Compare the work for (a) a reversible path and (b) a single-step expansion against a constant external pressure equal to the final gas pressure.
($R = 8.314\ \text{J mol}^{-1}\,\text{K}^{-1} = 0.08206\ \text{L atm mol}^{-1}\,\text{K}^{-1}$)

(a) $$w_{\text{rev}} = -2.303\,nRT\log_{10}\frac{V_2}{V_1} = -2.303 \times 1 \times 8.314 \times 300 \times \log_{10} 2$$
$$= -2.303 \times 2494.2 \times 0.3010 \approx \boxed{-1729\ \text{J}}$$

(b) $$P_2 = \frac{nRT}{V_2} = \frac{0.08206 \times 300}{20} = 1.231\ \text{atm}$$
$$w_{\text{irr}} = -1.231 \times (20 - 10) = -12.31\ \text{L atm} = -12.31 \times 101.3 \approx \boxed{-1247\ \text{J}}$$

Result: the reversible path gives the larger magnitude of work (1729 J > 1247 J), exactly as the graph predicts. In both cases $\Delta U = 0$, so the heat absorbed is $q = -w$.

◆ **BRIDGE TO THE NEXT IDEA**

So far we kept T fixed. What if we fix the volume or the pressure instead and ask how much heat changes the temperature? That brings us back to heat capacity, now with a constraint.

7 Calculating C_v and C_p

7.1 Constant-volume (isochoric) process

Start from the first law in differential form:

$$dU = dq + dw = dq - P\,dV$$

At constant volume $dV = 0$, so no P–V work is done and the work term vanishes: $dU = dq_v$. From the definition of molar heat capacity, $dq_v = nC_v\,dT$. Hence:

KEY RESULT

$$dU = dq_v = nC_v\,dT \quad \Longrightarrow \quad \Delta U = nC_v\,\Delta T$$

So at constant volume, **the heat absorbed equals the change in internal energy:** $q_v = \Delta U$.

◆ **EXTENSION FOR IDEAL GASES**

Since U of an ideal gas depends only on T, the expression $\Delta U = nC_v\,\Delta T$ holds for **any** process of an ideal gas, not only constant-volume ones.

7.2 Constant-pressure (isobaric) process

Now the volume can change, so the gas does P–V work. Starting again from $dU = dq - P\,dV$ and rearranging, with the heat now measured at constant pressure:

$$dU + P\,dV = dq_p = nC_p\,dT$$

The heat supplied is shared between raising the internal energy (dU) and doing expansion work ($P\,dV$). This is why C_p is expected to be **larger** than C_v for a gas. The combination $U + PV$ keeps appearing whenever heat is measured at constant pressure, so it deserves its own name and symbol.

◆ **BRIDGE TO THE NEXT IDEA**

Most laboratory chemistry happens in open flasks at constant atmospheric pressure, so we need a state function that equals the heat at constant pressure. That is **enthalpy**.

8 Enthalpy (H)

From the previous section: at constant volume $q_v = \Delta U$. At constant pressure, with only P–V work:

$$\Delta U = q_p - p\,\Delta V \quad\Longrightarrow\quad q_p = (U_2 + pV_2) - (U_1 + pV_1)$$

Define the new function **enthalpy** (from Greek *enthalpein*, “to warm”):

DEFINITION

$$H = U + pV \quad\Longrightarrow\quad q_p = \Delta H$$

- H is a **state function** because U, p and V are all state functions. Hence ΔH is path-independent, and so is q_p, even though q in general is a path function.
- **Exothermic** reactions release heat: ΔH is negative. **Endothermic** reactions absorb heat: ΔH is positive.
- At constant pressure: $\Delta H = \Delta U + p\,\Delta V$. At constant volume ($\Delta V = 0$): $q_v = \Delta U$; here $\Delta H = \Delta U + V\Delta p$, which is close to ΔU only when the pressure change is negligible (e.g. solids and liquids).

ΔH versus ΔU in reactions

For solids and liquids, volume barely changes on heating, so $\Delta H \approx \Delta U$. The difference matters when **gases** are involved. For gaseous reactants (n_A mol) and products (n_B mol) at constant T and p, the ideal gas law gives $pV_A = n_A RT$ and $pV_B = n_B RT$, so:

KEY RESULT

$$p\,\Delta V = \Delta n_g RT \quad \Longrightarrow \quad \Delta H = \Delta U + \Delta n_g RT$$

Here Δn_g = (moles of gaseous products) − (moles of gaseous reactants). This relation converts ΔH to ΔU and vice versa.

◆ **BRIDGE TO THE NEXT IDEA**

We now have $\Delta U = nC_v\,\Delta T$ and $\Delta H = nC_p\,\Delta T$. Since $H = U + pV$, the two heat capacities must be connected. Let us derive how.

9 Relationship Between C_p and C_v for an Ideal Gas

For one mole of an ideal gas, $pV = RT$, so:

$$\Delta H = \Delta U + \Delta(pV) = \Delta U + \Delta(RT) = \Delta U + R\,\Delta T$$

Substitute $\Delta H = C_p\,\Delta T$ and $\Delta U = C_v\,\Delta T$ (molar quantities):

KEY RESULT

$$C_p\,\Delta T = C_v\,\Delta T + R\,\Delta T \quad \Longrightarrow \quad \boldsymbol{C_p - C_v = R}$$

◆ **PHYSICAL MEANING**

At constant pressure, part of the heat supplied goes into raising the temperature (the C_v part) and the rest is spent as expansion work against the surroundings (the R part per mole). So $C_p > C_v$ for a gas.

◆ **QUICK NUMBERS**

For a monatomic ideal gas $C_v = \frac{3}{2}R \approx 12.47\ \text{J mol}^{-1}\,\text{K}^{-1}$, and so $C_p = \frac{5}{2}R \approx 20.79\ \text{J mol}^{-1}\,\text{K}^{-1}$.

Σ Summary: Formula Sheet

Idea	Formula	Remember
First law	$\Delta U = q + w$	$w > 0$ when work is done **on** the system
Heat capacity	$q = C\,\Delta T;\ C_{\mathrm{m}} = \dfrac{q}{n\Delta T};\ c = \dfrac{q}{m\Delta T}$	Process-dependent: ∞ isothermal, 0 adiabatic
P–V work	$w = -P_{\mathrm{ext}}\,\Delta V$	Expansion: $w < 0$. Compression: $w > 0$
Isothermal reversible	$w = -nRT\ln\dfrac{V_2}{V_1} = -2.303\,nRT\log_{10}\dfrac{V_2}{V_1}$	$\Delta U = 0$ for an ideal gas, so $q = -w$
Isothermal irreversible	$w = -P_{\mathrm{ext}}(V_2 - V_1)$	Expansion: $\lvert w_{\mathrm{rev}}\rvert > \lvert w_{\mathrm{irr}}\rvert$ Compression: $w_{\mathrm{irr}} > w_{\mathrm{rev}}$
Constant volume	$\Delta U = q_v = nC_v\,\Delta T$	$\Delta U = nC_v\,\Delta T$ for **any** ideal-gas process
Constant pressure	$\Delta H = q_p = nC_p\,\Delta T$	$H = U + pV$, a state function
ΔH and ΔU	$\Delta H = \Delta U + p\,\Delta V = \Delta U + \Delta n_g RT$	Δn_g counts gaseous moles only
Heat-capacity relation	$C_p - C_v = R$	Ideal gas, molar quantities
Unit conversion	$1\ \mathrm{L\,atm} = 101.3\ \mathrm{J}$	$R = 8.314\ \mathrm{J\,mol^{-1}\,K^{-1}}$